

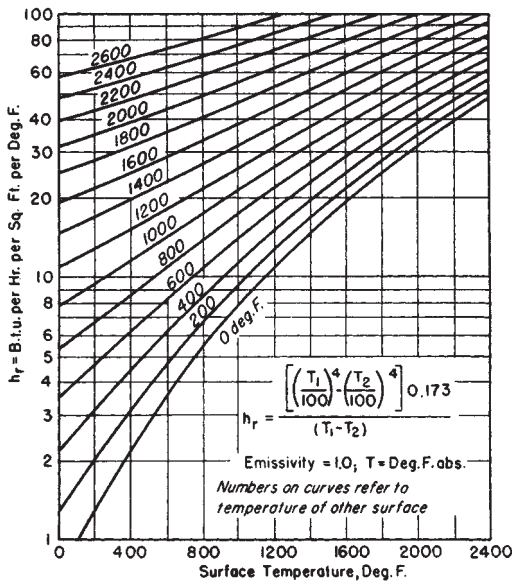


FIG. 5-7 Radiation coefficients of heat transfer h_r . To convert British thermal units per hour-square foot-degrees Fahrenheit to joules per square meter-second-kelvins, multiply by 5.6783; $^{\circ}\text{C} = (^{\circ}\text{F} - 32)/1.8$.

FORCED CONVECTION

Forced-convection heat transfer is the most frequently employed mode of heat transfer in the process industries. Hot and cold fluids, separated by a solid boundary, are pumped through the heat-transfer equipment, the rate of heat transfer being a function of the physical properties of the fluids, the flow rates, and the geometry of the system. Flow is generally turbulent, and the flow duct varies in complexity from circular tubes to baffled and extended-surface heat exchangers. Theoretical analyses of forced-convection heat transfer have been limited to relatively simple geometries and laminar flow. Analyses of turbulent-flow heat transfer have been based upon some mechanistic model and have not generally yielded relationships which were suitable for design purposes. Usually for complicated geometries only empirical relationships are available, and frequently these are based upon limited data and special operating conditions. Heat-transfer coefficients are strongly influenced by the mechanics of flow occurring during forced-convection heat transfer. Intensity of turbulence, entrance conditions, and wall conditions are some of the factors which must be considered in detail as greater accuracy in prediction of coefficients is required.

Analogy between Momentum and Heat Transfer The inter-relationship of momentum transfer and heat transfer is obvious from examining the equations of motion and energy. For constant fluid properties, the equations of motion must be solved before the energy equation is solved. If fluid properties are not constant, the equations are coupled, and their solutions must proceed simultaneously. Con-

siderable effort has been directed toward deriving some simple relationship between momentum and heat transfer. The methodology has been to use easily observed velocity profiles to obtain a measure of the diffusivity of momentum in the flowing stream. The analogy between heat and momentum is invoked by assuming that diffusion of heat and diffusion of momentum occur by essentially the same mechanism so that a relatively simple relationship exists between the diffusion coefficients. Thus, the diffusivity of momentum is used to predict temperature profiles and thence by Eq. (10-25) to predict the heat-transfer coefficient.

The analogy has been reasonably successful for simple geometries and for fluids of very low Prandtl number (liquid metals). For high-Prandtl-number fluids the **empirical analogy of Colburn** [*Trans. Am. Inst. Chem. Eng.*, **29**, 174 (1933)] has been very successful. A j factor for momentum transfer is defined as $j = f/2$, where f is the friction factor for the flow. The j factor for heat transfer is assumed to be equal to the j factor for momentum transfer

$$j = h/cG(c\mu/k)^{2/3} \tag{5-37}$$

More involved analyses for **circular tubes** reduce the equations of motion and energy to the form

$$\frac{\tau_{yz}}{\rho} = - \frac{(v + \epsilon_M)du}{dy} \tag{5-38a}$$

$$\frac{q/A}{c\rho} = - \frac{(\alpha + \epsilon_H)dt}{dy} \tag{5-38b}$$

where ϵ_H is the eddy diffusivity of heat and ϵ_M is the eddy diffusivity of momentum. The units of diffusivity are L^2/θ . The eddy viscosity is $E_M = \rho\epsilon_M$, and the eddy conductivity of heat is $E_H = \epsilon_H c\rho$. Values of ϵ_M are determined via Eq. (5-38a) from experimental velocity-distribution data. By assuming $\epsilon_H/\epsilon_M = \text{constant}$ (usually unity), Eq. (10-38b) is solved to give the temperature distribution from which the heat-transfer coefficient may be determined. The major difficulties in solving Eq. (5-38b) are in accurately defining the thickness of the various flow layers (laminar sublayer and buffer layer) and in obtaining a suitable relationship for prediction of the eddy diffusivities. For assistance in predicting eddy diffusivities, see Reichardt [NACA Tech. Memo 1408, 1957] and Strunk and Chao [*Am. Inst. Chem. Eng. J.*, **10**, 269 (1964)].

Internal and External Flow Two main types of flow are considered in this subsection: internal or conduit flow, in which the fluid completely fills a closed stationary duct, and external or immersed flow, in which the fluid flows past a stationary immersed solid. With **internal flow**, the heat-transfer coefficient is theoretically infinite at the location where heat transfer begins. The local heat-transfer coefficient rapidly decreases and becomes constant, so that after a certain length the average coefficient in the conduit is independent of the length. The local coefficient may follow an irregular pattern, however, if obstructions or turbulence promoters are present in the duct. For **immersed flow**, the local coefficient is again infinite at the point where heating begins, after which it decreases and may show various irregularities depending upon the configuration of the body. Usually in this instance the local coefficient never becomes constant as flow proceeds downstream over the body.

When heat transfer occurs during immersed flow, the rate is dependent upon the configuration of the body, the position of the body, the proximity of other bodies, and the flow rate and turbulence of the

TABLE 5-2 Values of $(h_c + h_r)^*$

Btu/(h-ft²·°F from pipe to room)
For horizontal bare standard steel pipe of various sizes in a room at 80°F

Nominal pipe diameter, in	Temperature difference, °F													
	30	50	100	150	200	250	300	350	400	450	500	550	600	700
1	2.16	2.26	2.50	2.73	3.00	3.29	3.60	3.95	4.34	4.73	5.16	5.60	6.05	6.98
3	1.97	2.05	2.25	2.47	2.73	3.00	3.31	3.69	4.03	4.43	4.85	5.26	5.71	6.66
5		1.95	2.15	2.36	2.61	2.90	3.20	3.54	3.90					
10	1.80	1.87	2.07	2.29	2.54	2.82	3.12	3.47	3.84					

*Bailey and Lyell [*Engineering*, **147**, 60 (1939)] give values for $(h_c + h_r)$ up to Δt , of 1000°F. $^{\circ}\text{C} = (^{\circ}\text{F} - 32)/1.8$; 5.6783 Btu/(h-ft²·°F) = J/(m²·s·K).

TABLE 5-3 Values of a and m for Eq. (5-32)

Configuration	$N_{Gr}N_{Pr}(\delta/L)^3$	a	m
Vertical spaces	2×10^4 to 2×10^5	$0.20 (\delta/L)^{-5/36}$	$1/4$
	2×10^5 to 10^7	$0.071 (\delta/L)^{1/9}$	$1/3$
Horizontal spaces	10^4 to 3×10^5	$0.21 (\delta/L)^{-1/4}$	$1/4$
	3×10^5 to 10^7	0.075	$1/3$

δ = cell width, L = cell length.

stream. The heat-transfer coefficient varies over the immersed body, since both the thermal and the momentum boundary layers vary in thickness. Relatively simple relationships are available for simple configurations immersed in an infinite flowing fluid. For complicated configurations and assemblages of bodies such as are found on the shell side of a heat exchanger, little is known about the local heat-transfer coefficient; empirical relationships giving average coefficients are all that are usually available. Research that has been conducted on local coefficients in complicated geometries has not been extensive enough to extrapolate into useful design relationships.

Laminar Flow Normally, laminar flow occurs in closed ducts when $N_{Re} < 2100$ (based on equivalent diameter $D_e = 4 \times \text{free area} \div \text{perimeter}$). Laminar-flow heat transfer has been subjected to extensive theoretical study. The energy equation has been solved for a variety of boundary conditions and geometrical configurations. However, true laminar-flow heat transfer very rarely occurs. Natural-convection effects are almost always present, so that the assumption that molecular conduction alone occurs is not valid. Therefore, empirically derived equations are most reliable.

Data are most frequently correlated by the Nusselt number $(N_{Nu})_{lm}$ or $(N_{Nu})_{am}$, the Graetz number $N_{Gz} = (N_{Re}N_{Pr}D/L)$, and the Grashof (natural-convection effects) number N_{Gr} . Some correlations consider only the variation of viscosity with temperature, while others also consider density variation. Theoretical analyses indicate that for very long tubes $(N_{Nu})_{lm}$ approaches a limiting value. Limiting Nusselt numbers for various closed ducts are shown in Table 5-4.

Circular Tubes For horizontal tubes and constant wall temperature, several relationships are available, depending on the Graetz number. For $0.1 < N_{Gz} < 10^4$, Hausen's [Allg. Waermetech., 9, 75 (1959)], the following equation is recommended.

$$(N_{Nu})_{lm} = 3.66 + \frac{0.19N_{Gz}^{0.8}}{1 + 0.117N_{Gz}^{0.467}} \left(\frac{\mu_b}{\mu_w} \right)^{0.14} \quad (5-39)$$

For $N_{Gz} > 100$, the Sieder-Tate relationship [Ind. Eng. Chem., 28, 1429 (1936)] is satisfactory for small diameters and Δt 's:

$$(N_{Nu})_{am} = 1.86N_{Gz}^{1/3} (\mu_b/\mu_w)^{0.14} \quad (5-40)$$

A more general expression covering all diameters and Δt 's is obtained by including an additional factor $0.87(1 + 0.015N_{Gz}^{1/3})$ on the right side of Eq. (5-40). The diameter should be used in evaluating N_{Gr} . An equation published by Oliver [Chem. Eng. Sci., 17, 335 (1962)] is also recommended.

TABLE 5-4 Values of Limiting Nusselt Number in Laminar Flow in Closed Ducts

Configuration	Limiting Nusselt number $N_{Gr} < 4.0$	
	Constant wall temperature	Constant heat flux
Circular tube	3.66	4.36
Concentric annulus		Eq. (10-42)
Equilateral triangle		3.00
Rectangles		
Aspect ratio:		
1.0 (square)	2.89	3.63
0.713		3.78
0.500	3.39	4.11
0.333		4.77
0.25		5.35
0 (parallel planes)	7.60	8.24

For laminar flow in vertical tubes a series of charts developed by Pigford [Chem. Eng. Prog. Symp. Ser. 17, 51, 79 (1955)] may be used to predict values of h_{ann} .

Annuli Approximate heat-transfer coefficients for laminar flow in annuli may be predicted by the equation of Chen, Hawkins, and Solberg [Trans. Am. Soc. Mech. Eng., 68, 99 (1946)]:

$$(N_{Nu})_{am} = 1.02N_{Re}^{0.45}N_{Pr}^{0.5} \left(\frac{D_e}{L} \right)^{0.4} \left(\frac{D_2}{D_1} \right)^{0.8} \left(\frac{\mu_b}{\mu_1} \right)^{0.14} N_{Gr}^{0.05} \quad (5-41)$$

Limiting Nusselt numbers for slug-flow annuli may be predicted (for constant heat flux) from Trefethen [General Discussions on Heat Transfer, London, ASME, New York, 1951, p. 436]:

$$(N_{Nu})_{lm} = \frac{8(m-1)(m^2-1)^2}{4m^4 \ln m - 3m^4 + 4m^2 - 1} \quad (5-42)$$

where $m = D_2/D_1$. The Nusselt and Reynolds numbers are based on the equivalent diameter, $D_2 - D_1$.

Limiting Nusselt numbers for laminar flow in annuli have been calculated by Dwyer [Nucl. Sci. Eng., 17, 336 (1963)]. In addition, theoretical analyses of laminar-flow heat transfer in concentric and eccentric annuli have been published by Reynolds, Lundberg, and McCuen [Int. J. Heat Mass Transfer, 6, 483, 495 (1963)]. Lee [Int. J. Heat Mass Transfer, 11, 509 (1968)] presented an analysis of turbulent heat transfer in entrance regions of concentric annuli. Fully developed local Nusselt numbers were generally attained within a region of 30 equivalent diameters for $0.1 < N_{Pr} < 30$, $10^4 < N_{Re} < 2 \times 10^5$, $1.01 < D_2/D_1 < 5.0$.

Parallel Plates and Rectangular Ducts The limiting Nusselt number for parallel plates and flat rectangular ducts is given in Table 5-4. Norris and Streid [Trans. Am. Soc. Mech. Eng., 62, 525 (1940)] report for constant wall temperature

$$(N_{Nu})_{lm} = 1.85N_{Gz}^{1/3} \quad (5-43)$$

for $N_{Gz} > 70$. Both Nusselt number and Graetz numbers are based on equivalent diameter. For large temperature differences it is advisable to apply the correction factor $(\mu_b/\mu_w)^{0.14}$ to the right side of Eq. (5-43).

For rectangular ducts Kays and Clark (Stanford Univ., Dept. Mech. Eng. Tech. Rep. 14, Aug. 6, 1953) published relationships for heating and cooling of air in rectangular ducts of various aspect ratios. For most noncircular ducts Eqs. (5-39) and (5-40) may be used if the equivalent diameter ($= 4 \times \text{free area/wetted perimeter}$) is used as the characteristic length. See also Kays and London, Compact Heat Exchangers, 3d ed., McGraw-Hill, New York, 1984.

Immersed Bodies When flow occurs over immersed bodies such that the boundary layer is completely laminar over the whole body, laminar flow is said to exist even though the flow in the mainstream is turbulent. The following relationships are applicable to single bodies immersed in an infinite fluid and are not valid for assemblages of bodies.

In general, the average heat-transfer coefficient on immersed bodies is predicted by

$$N_{Nu} = C_r(N_{Re})^m(N_{Pr})^{1/3} \quad (5-44)$$

Values of C_r and m for various configurations are listed in Table 5-5. The characteristic length is used in both the Nusselt and the Reynolds numbers, and the properties are evaluated at the film temperature $= (t_w + t_\infty)/2$. The velocity in the Reynolds number is the undisturbed free-stream velocity.

Heat transfer from immersed bodies is discussed in detail by Eckert and Drake, Jakob, and Knudsen and Katz (see "Introduction: General References"), where equations for local coefficients and the effects of unheated starting length are presented. Equation (5-44) may also be expressed as

$$N_{St}N_{Pr}^{2/3} = C_rN_{Re}^{m-1} = f/2 \quad (5-45)$$

where f is the skin-friction drag coefficient (not the form drag coefficient).

Falling Films When a liquid is distributed uniformly around the periphery at the top of a vertical tube (either inside or outside) and allowed to fall down the tube wall by the influence of gravity, the fluid

TABLE 5-5 Laminar-Flow Heat Transfer over Immersed Bodies [Eq. (5-44)]

Configuration	Characteristic length	N_{Re}	N_{Pr}	C_r	m
Flat plate parallel to flow Circular cylinder axes perpendicular to flow	Plate length	10^3 to 3×10^5	>0.6	0.648	0.50
	Cylinder diameter	1 – 4		0.989	0.330
		4 – 40		0.911	0.385
		40 – 4000	>0.6	0.683	0.466
		$4 \times 10^3 - 4 \times 10^4$		0.193	0.618
Non-circular cylinder, axis Perpendicular to flow, characteristic Length perpendicular to flow	Square, short diameter Square, long diameter Hexagon, short diameter Hexagon, long diameter	$4 \times 10^4 - 2.5 \times 10^5$		0.0266	0.805
		$5 \times 10^3 - 10^5$	>0.6	0.104	0.675
		$5 \times 10^3 - 10^5$		0.250	0.588
		$5 \times 10^3 - 10^5$		0.155	0.638
		$5 \times 10^3 - 2 \times 10^4$		0.162	0.638
		$2 \times 10^4 - 10^5$		0.0391	0.782
Sphere*	Diameter	$1 - 7 \times 10^4$	0.6 – 400	0.6	0.50

*Replace N_{Nu} by $N_{Nu} - 2.0$ in Eq. (5-44).

does not fill the tube but rather flows as a thin layer. Similarly, when a liquid is applied uniformly to the outside and top of a horizontal tube, it flows in layer form around the periphery and falls off the bottom. In both these cases the mechanism is called gravity flow of liquid layers or falling films.

For the turbulent flow of **water** in layer form down the walls of **vertical tubes** the dimensional equation of McAdams, Drew, and Bays [Trans. Am. Soc. Mech. Eng., **62**, 627 (1940)] is recommended:

$$h_{lm} = b \Gamma^{1/3} \tag{5-46}$$

where $b = 9150$ (SI) or 120 (U.S. customary) and is based on values of $\Gamma = W_p/\pi D$ ranging from 0.25 to 6.2 kg/(m·s) [600 to 15,000 lb/(h·ft)] of wetted perimeter. This type of water flow is used in vertical vapor-in-shell ammonia condensers, acid coolers, cycle water coolers, and other process-fluid coolers.

The following dimensional equations may be used for **any liquid** flowing in layer form down **vertical surfaces**:

For $\frac{4\Gamma}{\mu} > 2100$
$$h_{lm} = 0.01 \left(\frac{k^3 \rho^2 g}{\mu^2} \right)^{1/3} \left(\frac{c\mu}{k} \right)^{1/3} \left(\frac{4\Gamma}{\mu} \right)^{1/3} \tag{5-47a}$$

For $\frac{4\Gamma}{\mu} < 2100$
$$h_{am} = 0.50 \left(\frac{k^3 \rho^{4/3} c_g^{2/3}}{L \mu^{1/3}} \right)^{1/3} \left(\frac{\mu}{\mu_w} \right)^{1/4} \left(\frac{4\Gamma}{\mu} \right)^{1/9} \tag{5-47b}$$

Equation (5-47b) is based on the work of Bays and McAdams [Ind. Eng. Chem., **29**, 1240 (1937)]. The significance of the term L is not clear. When $L = 0$, the coefficient is definitely not infinite. When L is large and the fluid temperature has not yet closely approached the wall temperature, it does not appear that the coefficient should necessarily decrease. Within the finite limits of 0.12 to 1.8 m (0.4 to 6 ft), this equation should give results of the proper order of magnitude.

For falling films applied to the **outside of horizontal tubes**, the Reynolds number rarely exceeds 2100. Equations may be used for falling films on the outside of the tubes by substituting $\pi D/2$ for L .

For **water** flowing over a **horizontal tube**, data for several sizes of pipe are roughly correlated by the dimensional equation of McAdams, Drew, and Bays [Trans. Am. Soc. Mech. Eng., **62**, 627 (1940)].

$$h_{am} = b (\Gamma/D_0)^{1/3} \tag{5-48}$$

where $b = 3360$ (SI) or 65.6 (U.S. customary) and Γ ranges from 0.94 to 4 kg/(m·s) [100 to 1000 lb/(h·ft)].

Falling films are also used for evaporation in which the film is both entirely or partially evaporated (juice concentration). This principle is also used in crystallization (freezing).

The advantage of high coefficient in falling-film exchangers is partially offset by the difficulties involved in distribution of the film, maintaining complete wettability of the tube, and pumping costs required to lift the liquid to the top of the exchanger.

Transition Region Turbulent-flow equations for predicting heat transfer coefficients are usually valid only at Reynolds numbers greater than 10,000. The transition region lies in the range $2000 < N_{Re} < 10,000$. No simple equation exists for accomplishing a smooth mathematical transition from laminar flow to turbulent flow. Of the relationships proposed, Hausen's equation [Z. Ver. Dtsch. Ing. Beih. Verfahrenstechn., No.

4, 91 (1934)] fits both the laminar extreme and the fully turbulent extreme quite well.

$$(N_{Nu})_{lam} = 0.116(N_{Re}^{2/3} - 125)N_{Pr}^{1/3} \left[1 + \left(\frac{D}{L} \right)^{2/3} \right] \left(\frac{\mu_b}{\mu_w} \right)^{0.14} \tag{5-49}$$

between 2100 and 10,000. It is customary to represent the probable magnitude of coefficients in this region by hand-drawn curves (Fig. 5-8). Equation (5-40) is plotted as a series of curves (j factor versus Reynolds number with L/D as parameters) terminating at Reynolds number = 2100. Continuous curves for various values of L/D are then hand-drawn from these terminal points to coincide tangentially with the curve for forced-convection, fully turbulent flow [Eq. (5-50c)].

Turbulent Flow

Circular Tubes Numerous relationships have been proposed for predicting turbulent flow in tubes. For high-Prandtl-number fluids, relationships derived from the equations of motion and energy through the momentum-heat-transfer analogy are more complicated and no more accurate than many of the empirical relationships that have been developed.

For $N_{Re} > 10,000$, $0.7 < N_{Pr} < 170$, for properties based on the bulk temperature and for **heating**, the **Dittus-Boelter equation** [Boelter, Cherry, Johnson and Martinelli, Heat Transfer Notes, McGraw-Hill, New York (1965)] may be used:

$$N_{Nu} = 0.0243 N_{Re}^{0.8} N_{Pr}^{0.4} (\mu_b/\mu_w)^{0.14} \tag{5-50a}$$

For **cooling**, the relationship is

$$N_{Nu} = 0.0265 N_{Re}^{0.8} N_{Pr}^{0.3} (\mu_b/\mu_w)^{0.14} \tag{5.50b}$$

The **Colburn correlation** is

$$j_H = N_{St} N_{Pr}^{2/3} (\mu_w/\mu_b)^{0.14} = 0.023 N_{Re}^{-0.2} \tag{5-50c}$$

In Eq. (5-50c), the viscosity-ratio factor may be neglected if properties are evaluated at the film temperature $(t_b + t_w)/2$.

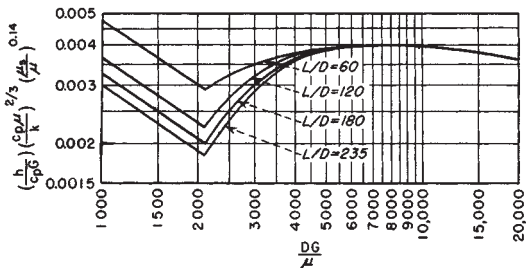


FIG. 5-8 Graphical representation of the Colburn j factor for the heating and cooling of fluids inside tubes. The curves for N_{Re} below 2100 are based on Eq. (5-40). L is the length of each pass in feet. The curve for N_{Re} above 10,000 is represented by Eq. (5-50c).

For the **transition and turbulent regions, including diameter to length effects**, Gnielinski [*Int. Chem. Eng.*, **16**, 359 (1976)] recommends a modification of an equation suggested by Petukhov and Popov [*High Temp.*, **1**, 69 (1963)]. This equation applies in the ranges $0 < D/L < 1$, $0.6 < N_{\text{Re}} < 2000$, $2300 < N_{\text{Re}} < 10^6$.

$$N_{\text{Nu}} = \frac{(f/2)(N_{\text{Re}} - 1000) N_{\text{Pr}}}{1 + 12.7(f/2)^{0.5}(N_{\text{Pr}}^{2/3} - 1)} \left(1 + \left(\frac{D}{L} \right)^{2/3} \right) \left(\frac{\mu_b}{\mu_w} \right)^{0.14} \quad (5-51a)$$

The Fanning friction f is determined by an equation recommended by Filonenko [*Teploenergetika*, **1**, 40 (1954)]

$$f = 0.25 (1.82 \log_{10} N_{\text{Re}} - 1.64)^{-2} \quad (5-51b)$$

Any other appropriate friction factor equation for smooth tubes may be used.

Approximate predictions for **rough pipes** may be obtained from Eq. (5-50c) if the right-hand term is replaced by $f/2$ for the rough pipe. For air, Nunner (*Z. Ver. Dtsch. Ing. Forsch.*, 1956, p. 455) obtains

$$\frac{(N_{\text{Nu}})_{\text{rough}}}{(N_{\text{Nu}})_{\text{smooth}}} = \frac{f_{\text{rough}}}{f_{\text{smooth}}} \quad (5-52)$$

Dippery and Sabersky [*Int. J. Heat Mass Transfer*, **6**, 329 (1963)] present a complete discussion of the influence of roughness on heat transfer in tubes.

Dimensional Equations for Various Conditions For gases at ordinary pressures and temperatures based on $c\mu/k = 0.78$ and $\mu = (1.76)(10^{-5})$ Pa·s [0.0426 lb/(ft·h)]

$$h = b c p^{0.8} (V^{0.8}/D^{0.2}) \quad (5-53)$$

where $b = (3.04)(10^{-3})$ (SI) or $(1.44)(10^{-2})$ (U.S. customary). For air at atmospheric pressure

$$h = b (V^{0.8}/D^{0.2}) \quad (5-54)$$

where $b = 3.52$ (SI) or $(4.35)(10^{-4})$ (U.S. customary). For water [based on a temperature range of 5 to 104°C (40 to 220°F)]

$$h = 1057 (1.352 + 0.02t) (V^{0.8}/D^{0.2}) \quad (5-55a)$$

in SI units with $t = ^\circ\text{C}$, or

$$h = 0.13(1 + 0.011t)(V^{0.8}/D^{0.2}) \quad (5-55b)$$

in U.S. customary units with $t = ^\circ\text{F}$.

For organic liquids, based on $c = 2.092$ J/kg·K [0.5 Btu/(lb·°F)], $k = 0.14$ J/(m·s·K) [0.08 Btu/(h·ft·°F)], $\mu_b = (1)(10^{-3})$ Pa·s (1.0 cP), and $\rho = 810$ kg/m³ (50 lb/ft³),

$$h = b (V^{0.8}/D^{0.2}) \quad (5-56)$$

where $b = 423$ (SI) or $(5.22)(10^{-2})$ (U.S. customary). Within reasonable limits, coefficients for organic liquids are about one-third of the values obtained for water.

Entrance effects are usually not significant industrially if $L/D > 60$. Below this limit Nusselt recommended the conservative equation for $10 < L/D < 400$ and properties evaluated at bulk temperature

$$N_{\text{Nu}} = 0.036 N_{\text{Re}}^{0.8} N_{\text{Pr}}^{1/3} (L/D)^{-0.054} \quad (5-57)$$

It is common to correlate entrance effects by the equation

$$h_m/h = 1 + F(D/L) \quad (5-58)$$

where h is predicted by Eq. (5-50a) or (5-50b), and h_m is the mean coefficient for the pipe in question. Values of F are reported by Boelter, Young, and Iverson [NACA Tech. Note 1451, 1948] and tabulated by Kays and Knudsen and Katz (see "Introduction: General References"). Selected values of F are as follows:

Fully developed velocity profile	1.4
Abrupt contraction entrance	6
90° right-angle bend	7
180° round bend	6

For **large temperature differences** different equations are necessary and usually are specifically applicable to either gases or liquids. Gambill (*Chem. Eng.*, Aug. 28, 1967, p. 147) provides a detailed review of high-flux heat transfers to gases. He recommends

$$N_{\text{Nu}} = \frac{0.021 N_{\text{Re}}^{0.8} N_{\text{Pr}}^{0.4}}{(T_w/T_b)^{0.29 + 0.0019(L/D)}} \quad (5-59)$$

for $10 < L/D < 240$, $110 < T_b < 1560$ K ($200 < T_b < 2800^\circ\text{R}$), $1.1 < (T_w/T_b) < 8.0$, and properties evaluated at T_b . For liquids, Eq. (5-50c) is generally satisfactory.

Annuli For diameter ratios $D_2/D_1 > 0.2$, Monrad and Pelton's equation [*Trans. Am. Inst. Chem. Eng.*, **38**, 593 (1942)] is recommended for either or both the inner and outer tube:

$$N_{\text{Nu}} = 0.020 N_{\text{Re}}^{0.8} N_{\text{Pr}}^{1/3} (D_2/D_1)^{0.53} \quad (5-60a)$$

Equation (5-51a) may also be used for **smooth annuli** as follows:

$$\frac{(N_{\text{Nu}})_{\text{ann}}}{(N_{\text{Nu}})_{\text{tube}}} = \phi \left(\frac{D_1}{D_2} \right) \quad (5-60b)$$

The hydraulic diameter $D_2 - D_1$ is used in N_{Nu} , N_{Re} , and D/L is used for the annulus. The function on the right of Eq. (5-60b) is given by Petukhov and Roizen [*High Temp.*, **2**, 65 (1964)] as follows:

$$\begin{array}{ll} \text{Inner tube heated} & 0.86 (D_1/D_2)^{-0.16} \\ \text{Outer tube heated} & 1 - 0.14 (D_1/D_2)^{0.6} \end{array}$$

If both tubes are heated, the function is the sum of the above two functions divided by $1 + D_1/D_2$ [Stephan, *Chem. Ing. Tech.*, **34**, 207 (1962)]. The Colburn form of relationship may be employed for the individual walls of the annulus by using the individual friction factor for each wall [see Knudsen, *Am. Inst. Chem. Eng. J.*, **8**, 566 (1962)]:

$$j_{m1} = (N_{\text{St}})_1 N_{\text{Pr}}^{2/3} = f_1/2 \quad (5-61a)$$

$$j_{m2} = (N_{\text{St}})_2 N_{\text{Pr}}^{2/3} = f_2/2 \quad (5-61b)$$

Rothfus, Monrad, Sikchi, and Heideger [*Ind. Eng. Chem.*, **47**, 913 (1955)] report that the friction factor f_2 for the outer wall bears the same relation to the Reynolds number for the outer portion of the annular stream $2(r_2^2 - \lambda_m)/r_2 \mu$ as the friction factor for circular tubes does to the Reynolds number for circular tubes, where r_2 is the radius of the outer tube and λ_m is the position of maximum velocity in the annulus, estimated from

$$\lambda_m = \frac{r_2^2 - r_1^2}{\ln(r_2/r_1)^2} \quad (5-62)^*$$

To calculate the friction factor f_1 for the inner tube use the relation

$$f_1 = \frac{f_2 r_2 (\lambda_m - r_1^2)}{r_1 (r_2^2 - \lambda_m)} \quad (5-63)$$

There have been several analyses of turbulent heat transfer in annuli: for example, Deissler and Taylor (NACA Tech. Note 3451, 1955), Kays and Leung [*Int. J. Heat Mass Transfer*, **6**, 537 (1963)], Lee [*Int. J. Heat Transfer*, **11**, 509 (1968)], Sparrow, Hallman and Siegel [*Appl. Sci. Res.*, **7A**, 37 (1958)], and Johnson and Sparrow [*Am. Soc. Mech. Eng. J. Heat Transfer*, **88**, 502 (1966)]. The reader is referred to these for details of the analyses.

For **annuli containing externally finned tubes** the heat-transfer coefficients are a function of the fin configurations. Knudsen and Katz (*Fluid Dynamics and Heat Transfer*, McGraw-Hill, New York, 1958) present relationships for transverse finned tubes, spined tubes, and longitudinal finned tubes in annuli.

Noncircular Ducts Equations (5-50a) and (5-50b) may be employed for noncircular ducts by using the equivalent diameter $D_e = 4 \times \text{free area per wetted perimeter}$. Kays and London (*Compact Heat Exchangers*, 3rd ed., McGraw-Hill, New York, 1984) give charts for various noncircular ducts encountered in compact heat exchangers.

Vibrations and pulsations generally tend to increase heat-transfer coefficients.

Example 2: Calculation of j Factors in an Annulus Calculate the heat-transfer j factors for both walls of an annulus for the following condi-

* Equation (5-62) predicts the point of maximum velocity for laminar flow in annuli and is only an approximate equation for turbulent flow. Brighton and Jones [*Am. Soc. Mech. Eng. Basic Eng.*, **86**, 835 (1964)] and Macagno and McDougall [*Am. Inst. Chem. Eng. J.*, **12**, 437 (1966)] give more accurate equations for predicting the point of maximum velocity for turbulent flow.

tions: $D_1 = 0.0254$ m (1.0 in); $D_2 = 0.0635$ m (2.5 in); water at 15.6°C (60°F); $\mu/\rho = (1.124)(10^{-6})$ m²/s [$(1.21)(10^{-5})$ ft²/s]; velocity = 1.22 m/s (4 ft/s).

$$\lambda_m = \frac{0.0635^2 - 0.0254^2}{4 \ln(0.0635/0.0254)} = (4.621)(10^{-4}) \text{ m} \quad (0.716 \text{ in})$$

$$\text{Re}_2 = \frac{2(r_2^2 - \lambda_m)V_p}{r_2\mu} = \frac{2[0.0318^2 - (4.621)(10^{-4})(1.22)]}{(0.0318)(1.124)(10^{-6})} = (3.74)(10^4)$$

From Eq. (5-51b), $f_2 = 0.0055$. Hence

$$j_{H2} = (N_{St})_2 N_{Pr}^{2/3} = 0.00275$$

From Eq. (5-63),

$$f_1 = \frac{(0.0055)(0.0318)[(4.621)(10^{-4}) - 0.0127^2]}{(0.0127)(0.0318^2 - (4.621)(10^{-4}))} = 0.00754$$

from which $j_{H1} = (N_{St})_1 N_{Pr}^{2/3} = 0.00377$.

These results indicate that for this system the heat-transfer coefficient on the inner tube is about 40 percent greater than on the outer tube.

Coils For flow *inside helical coils*, Reynolds number above 10,000, multiply the value of the film coefficient obtained from the applicable equation for straight tubes by the term $(1 + 3.5 D_i/D_c)$.

For flow *inside helical coils*, Reynolds number less than 10,000, substitute the term $(D_c/D_i)^{1/2}$ for (L/D_i) where the latter appears in the applicable equation for straight tubes (frequently as part of the Graetz number).

For **flat spiral (pancake) coils**, in which the ratio D_c/D_i varies for each turn, a different value of coefficient will be obtained for each turn; a weighted average based on length per turn is used.

For flow *outside helical coils* use the equation for flow normal to a bank of coils, in-line flow.

Finned Tubes (Extended Surface) When the film coefficient on the outside of a metal tube is much lower than that on the inside, as when steam condensing in a pipe is being used to heat air, externally finned (or extended) heating surfaces are of value in increasing substantially the rate of heat transfer per unit length of tube. The data on extended heating surfaces, for the case of air flowing outside and at right angles to the axes of a bank of finned pipes, can be represented approximately by the dimensional equation derived from

$$h_f = b \frac{V_F^{0.6}}{D_0^{0.4}} \left(\frac{p'}{p' - D_0} \right)^{0.6} \quad (5-64)$$

where $b = 5.29$ (SI) or $(5.39)(10^{-3})$ (U.S. customary); h_f is the film coefficient of heat transfer on the air side; V_F is the face velocity of the air; p' is the center-to-center spacing, m, of the tubes in a row; and D_0 is the outside diameter, m, of the bare tube (diameter at the root of the fins).

In atmospheric air-cooled finned tube exchangers, the air-film coefficient from Eq. (5-64) is sometimes converted to a value based on outside bare surface as follows:

$$h_{fo} = h_f \frac{A_f + A_{uf}}{A_{of}} = h_f \frac{A_f}{A_o} \quad (5-65)$$

in which h_{fo} is the air-film coefficient based on external bare surface; h_f is the air-film coefficient based on total external surface; A_f is total external surface, and A_o is external bare surface of the unfinned tube; A_f is the area of the fins; A_{uf} is the external area of the unfinned portion of the tube; and A_{of} is area of tube before fins are attached.

Fin efficiency is defined as the ratio of the mean temperature difference from surface to fluid divided by the temperature difference from fin to fluid at the base or root of the fin. Graphs of fin efficiency for extended surfaces of various types are given by Gardner [*Trans. Am. Soc. Mech. Eng.*, **67**, 621 (1945)].

Heat-transfer coefficients for finned tubes of various types are given in a series of papers [*Trans. Am. Soc. Mech. Eng.*, **67**, 601 (1945)].

For flow of air normal to fins in the form of **short strips or pins**, Norris and Spofford [*Trans. Am. Soc. Mech. Eng.*, **64**, 489 (1942)] correlate their results for air by the dimensionless equation of Pohlhausen:

$$\frac{h_m}{c_p G_{\max}} \left(\frac{c_p \mu}{k} \right)^{2/3} = 1.0 \left(\frac{z_p G_{\max}}{\mu} \right)^{-0.5} \quad (5-66)$$

for values of $z_p G_{\max}/\mu$ ranging from 2700 to 10,000.

For the general case, the treatment suggested by Kern (*Process Heat Transfer*, McGraw-Hill, New York, 1950, p. 512) is recommended. Because of the wide variations in fin-tube construction, it is convenient to convert all film coefficients to values based on the inside bare surface of the tube. Thus to convert the film coefficient based on outside area (finned side) to a value based on inside area Kern gives the following relationship:

$$h_{fi} = (\Omega A_f + A_o)(h_f/A_i) \quad (5-67)$$

in which h_{fi} is the effective outside film coefficient based on the inside area, h_f is the outside film coefficient calculated from the applicable equation for bare tubes, A_f is the surface area of the fins, A_o is the surface area on the outside of the tube which is not finned, A_i is the inside area of the tube, and Ω is the fin efficiency defined as

$$\Omega = (\tanh mb_f)/mb_f \quad (5-68)$$

in which

$$m = (h_f p_f / ka_s)^{1/2} \text{ m}^{-1} \text{ (ft}^{-1}\text{)} \quad (5-69)$$

and b_f = height of fin. The other symbols are defined as follows: p_f is the perimeter of the fin, a_s is the cross-sectional area of the fin, and k is the thermal conductivity of the material from which the fin is made.

Fin efficiencies and fin dimensions are available from manufacturers. Ratios of finned to inside surface are usually available so that the terms A_f , A_o , and A_i may be obtained from these ratios rather than from the total surface areas of the heat exchangers.

Banks of Tubes For heating and cooling of fluids flowing normal to a bank of circular tubes at least 10 rows deep the following equations are applicable:

Colburn type:

$$\frac{h}{c G_{\max}} \left(\frac{c \mu}{k} \right)^{2/3} = \frac{a}{(D_o G_{\max} / \mu)^{0.4}} = j \quad (5-70)$$

Nusselt type:

$$\frac{hD}{k} = a \left(\frac{D_o G_{\max}}{\mu} \right)^{0.6} \left(\frac{c \mu}{k} \right)^{1/3} \quad (5-71)$$

The dimensionless constant a in these equations varies depending upon conditions.

Conditions, Reynolds number > 3000	Value of a
Flow normal to apex of diamond, staggered arrangement	
No leakage	0.330
Normal leakage in baffled exchanger	0.198
Flow normal to flat side of diamond, not staggered (in-line) arrangement	
No leakage	0.260
Normal leakage in baffled exchanger	0.156

For Reynolds number less than 3000, Eq. (5-70) would give conservative results, but greater accuracy (if desired) may be obtained by using the following equation.

$$\frac{h}{c G_{\max}} \left(\frac{c \mu}{k} \right)^{2/3} = \frac{a}{(D_o G_{\max} / \mu)^m} = j \quad (5-72)$$

in which the constant a and exponent m are as follows:

Reynolds number	m	Tube pitch	Leakage	a
100–300	0.492	Staggered	None	0.695
			Normal	0.416
		In-line	None	0.548
			Normal	0.329
1–100	0.590	Staggered	None	1.086
			Normal	0.650
		In-line	None	0.855
			Normal	0.513

The following **dimensional equations** (5-73 to 5-77) are based on flow normal to a bank of staggered tubes without leakage. Multiply the values obtained for h by 0.6 for normal leakage and, in addition, by 0.79 for in-line (not staggered) tube arrangement.

$$h = b \frac{c^{1/3} k^{2/3} \rho^{0.6} \sqrt{v_{\max}^{0.6}}}{\mu^{0.267} D_0^{0.4}} \quad (5-73)$$

where $b = 0.33$ (SI) or 0.261 (U.S. customary). For gases at ordinary pressures and temperatures, based on $c\mu/k = 0.78$; $\mu = (1.76)(10^{-5})$ Pa·s [0.0426 lb/(ft·h)],

$$h = bc \frac{C_{\sigma_{\max}}^{0.6}}{D_0^{0.4}} \quad (5-74)$$

where $b = (4.82)(10^{-3})$ (SI) or 0.109 (U.S. customary). For air at atmospheric pressure

$$h = b \frac{V_{\max}^{0.6}}{D_0^{0.4}} \quad (5-75)$$

where $b = 5.33$ (SI) or $(5.44)(10^{-3})$ (U.S. customary). For water based on a temperature range 7 to 104°C (40 to 220°F)

$$h = 986(1.21 + 0.0121t) \frac{V_{\max}^{0.6}}{D_0^{0.4}} \quad (5-76a)$$

in SI units and t in °C.

$$h = 1.01(1 + 0.0067t) \frac{V_{\max}^{0.6}}{D_0^{0.4}} \quad (5-76b)$$

in U.S. customary units and t in °F. For organic liquids, based on $c = 2.22$ J/(kg·K) [0.53 Btu/(lb·°F)], $k = 0.14$ J/(m·s·K) [0.08 Btu/(h·ft·°F)], $\mu_b = (1)(10^{-3})$ Pa·s (1.0 cP), $\rho = 810$ kg/m³ (50 lb/ft³),

$$h = b \frac{V_{\max}^{0.6}}{D_0^{0.4}} \quad (5-77)$$

where $b = 400$ (SI) or 0.408 (U.S. customary).

JACKETS AND COILS OF AGITATED VESSELS

See Sec. 18.

NONNEWTONIAN FLUIDS

A wide variety of nonnewtonian fluids are encountered industrially. They may exhibit Bingham-plastic, pseudoplastic, or dilatant behavior and may or may not be thixotropic. For design of equipment to handle or process nonnewtonian fluids, the properties must usually be measured experimentally, since no generalized relationships exist to predict the properties or behavior of the fluids. Details of handling nonnewtonian fluids are described completely by Skelland (*Non-Newtonian Flow and Heat Transfer*; Wiley, New York, 1967). The generalized shear-stress rate-of-strain relationship for nonnewtonian fluids is given as

$$n' = \frac{d \ln (D \Delta P / 4L)}{d \ln (8V/D)} \quad (5-78)$$

as determined from a plot of shear stress versus velocity gradient.

For **circular tubes**, $N_{Gz} > 100$, $n' > 0.1$, and laminar flow

$$(N_{Nu})_{lm} = 1.75 \delta_s^{1/3} N_{Gz}^{1/3} \quad (5-79)$$

where $\delta_s = (3n' + 1)/4n'$. When natural-convection effects are considered, Metzner and Gluck [*Chem. Eng. Sci.*, **12**, 185 (1960)] obtained the following for **horizontal tubes**:

$$(N_{Nu})_{lm} = 1.75 \delta_s^{1/3} \left[N_{Gz} + 12.6 \left(\frac{N_{Pr} N_{Gr} D}{L} \right)^{0.4} \right]^{1/3} \left(\frac{\gamma_b}{\gamma_w} \right)^{0.14} \quad (5-80)$$

where properties are evaluated at the wall temperature, i.e., $\gamma = \gamma_w K'^{8n'-1}$ and $\tau_w = K'(8V/D)^{n'}$.

Metzner and Friend [*Ind. Eng. Chem.*, **51**, 879 (1959)] present relationships for turbulent heat transfer with nonnewtonian fluids. Relationships for heat transfer by natural convection and through laminar boundary layers are available in Skelland's book (op. cit.).

LIQUID METALS

Liquid metals constitute a class of heat-transfer media having Prandtl numbers generally below 0.01. Heat-transfer coefficients for liquid metals cannot be predicted by the usual design equations applicable to gases, water, and more viscous fluids with Prandtl numbers greater than 0.6. Relationships for predicting heat-transfer coefficients for liquid metals have been derived from solution of Eqs. (5-38a) and (5-38b). By the momentum-transfer-heat-transfer analogy, the eddy conductivity of heat is $kN_{Pr}(E_M/\mu) \approx k$ for small N_{Pr} . Thus in the solution of Eqs. (5-38a) and (5-38b) the knowledge of the thickness of various layers of flow is not critical. In fact, assumption of slug flow and constant conductivity ($=k$) across the duct gives reasonable values of heat-transfer coefficients for liquid metals.

For **constant heat flux**:

$$N_{Nu} = 5 + 0.025(N_{Re} N_{Pr})^{0.8} \quad (5-81)$$

For **constant wall temperature**:

$$N_{Nu} = 7 + 0.025(N_{Re} N_{Pr})^{0.8} \quad (5-82)$$

For $0.003 < N_{Pr} < 0.05$ and constant heat flux, Sleicher and Rouse [*Int. J. Heat Mass Transfer*, **18**, 677 (1975)] obtained the correlation

$$N_{Nu} = 6.3 + 0.0167 N_{Re}^{0.85} N_{Pr}^{0.93} \quad (5-83)$$

For **parallel plates and annuli** with $D_2/D_1 < 1.4$ and uniform heat flux, Seban [*Trans. Am. Soc. Mech. Eng.*, **72**, 789 (1950)] obtained the equation

$$N_{Nu} = 5.8 + 0.020(N_{Re} N_{Pr})^{0.8} \quad (5-84)$$

For annuli only, application of a factor of $0.70(D_2/D_1)^{0.53}$ is recommended for Eqs. (5-81) and (5-82). For more accurate semiempirical relationships for tubes, annuli, and rod bundles, refer to Dwyer [*Am. Inst. Chem. Eng. J.*, **9**, 261 (1963)].

Hsu [*Int. J. Heat Mass Transfer*, **7**, 431 (1964)] and Kalish and Dwyer [*Int. J. Heat Mass Transfer*, **10**, 1533 (1967)] discuss heat transfer to liquid metals flowing across **banks of tubes**. Hsu recommends the equations

$$N_{Nu} = 0.81 N_{Re} N_{Pr} (\phi/D)^{1/2} \quad (\text{for uniform heat flux}) \quad (5-85)$$

$$N_{Nu} = 0.096 N_{Re} N_{Pr} (\phi/D)^{1/2} \quad (\text{for cosine surface temperature}) \quad (5-86)$$

where the heat-transfer coefficient is based on the average circumferential temperature around the tubes, the Reynolds number is based on the superficial velocity through the tube bank, D is the tube outside diameter, and ϕ is a velocity potential function having the following values:

D/p'	ϕ/D square pitch	ϕ/D equilateral triangular pitch
0	2.00	2.00
0.1	2.02	2.02
0.2	2.07	2.06
0.3	2.16	2.15
0.4	2.30	2.27
0.5	2.52	2.45
0.6	2.84	2.71
0.7	3.34	3.11
0.8	4.23	3.80

Equations (5-85) and (5-86) are useful in calculating tube-surface temperatures.

Further information on liquid-metal heat transfer in tube banks is given by Hsu for spheres and elliptical rod bundles [*Int. J. Heat Mass Transfer*, **8**, 303 (1965)] and by Kalish and Dwyer for oblique flow across tube banks [*Int. J. Heat Mass Transfer*, **10**, 1533 (1967)]. For additional details of heat transfer with liquid metals for various systems see Dwyer (1968 ed., Na and Nak supplement to *Liquid Metals Handbook*) and Stein ("Liquid Metal Heat Transfer," in *Advances in Heat Transfer*, vol. 3, Academic, New York, 1966).